

ERDEN ERIM AYDOGDU

AI Engineer - LLM & RAG Systems - Financial AI - Enterprise Deployment

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PROFILE

AI Engineer specialising in LLM deployment, RAG architectures, and enterprise-scale AI systems. At Kentas (Uskudar Municipality R&D arm), deployed an institution-wide local LLM inference system from prototype to production and built multi-LLM pipelines for structured extraction from complex document corpora — promoted to the 3rd tier within 5 months. Experienced in end-to-end AI system design: embedding pipelines, vector database indexing, multi-agent orchestration, fine-tuning data workflows, and production-grade API security. Maintains 30+ active projects across AI/LLM, GIS & urban tech, full-stack, and DevTools. Background in technology consulting (CONSULTA) and delivery to senior institutional stakeholders — well-positioned for client-facing AI project delivery in the financial services sector.

TECHNICAL SKILLS

AI & LLM	RAG architecture, LangChain, LlamaIndex, prompt engineering, multi-agent systems (MCP), fine-tuning (LoRA, SFT), vector databases (pgvector, ChromaDB, Pinecone), embeddings, vLLM (GPU inference), Ollama, Gemini, OpenAI-compatible APIs, Groq
ML & Data	XGBoost, LightGBM, CatBoost, Optuna, CNN, ANN, PyTorch, structured JSON extraction, data pipeline design, scikit-learn
Full-Stack	React 19, Next.js 16, TypeScript, Node.js, FastAPI, PostgreSQL, Prisma, Tailwind CSS, WebSockets, Socket.io, RESTful API design
Spatial & GIS	PostGIS, MapLibre, CesiumJS, Deck.gl, ArcGIS, Leaflet, GeoAlchemy2, Turf.js, WebXR, Three.js, React Three Fiber
DevOps & Cloud	Docker, Kubernetes, CI/CD (GitHub Actions), Linux, Nginx, Git, Vercel, Railway, cloud-ready containerised deployments
AI Infra	vLLM (GPU inference), Ollama, ARQ async queue, Redis, better-sqlite3, XGBoost, LightGBM, CatBoost, Optuna
Hardware	STM32, IoT/MQTT, Embedded Systems, LTspice, PCB principles

PROFESSIONAL EXPERIENCE

Kentas - Uskudar Belediyesi R&D | Electrical & Electronics Engineer (Software & Systems) *Sep 2025 - Present*

Promoted to the 3rd tier within 5 months

- > Deployed Kentas AI Asistani: institution-wide local LLM inference system (vLLM + WSL2 + GPU) with custom system instruction design, stress-tested for production load, API security layer, and autonomous fine-tuning data pipeline capturing and labelling query-response pairs for future model retraining.
- > Built a multi-layered document AI pipeline — OCR ingestion of multiple PDFs, text embeddings, pgvector semantic indexing, MCP-based multi-agent orchestration — reducing institutional document retrieval from hours to under a minute.
- > Built MuteahhitHub, a CesiumJS Digital Twin platform integrating drone-captured 3D Tiles and GLB/GLTF models for real-time monitoring of 100+ active projects, used daily by senior management; live at mportal.uskudarkentas.com.tr.
- > Developed MBS & App Factory, a dynamic GIS application platform with zero-code map app generation, LDAP/JWT authentication, and RBAC — enabling municipal teams to deploy spatial tools without engineering involvement.
- > Built Sizinle Uskudar V2 (multi-tenant PPGIS), a citizen-facing platform serving 500,000+ with customisable admin panel, geographic tagging, moderation workflows, and rate limiting — rewritten from scratch for security and scalability.
- > Engineered a cross-platform XR showroom ecosystem in Unreal Engine (PC/VR) and WebXR (AR/VR) — consistent interactive experience across 4 platforms from VR headset to mobile browser.

Kafein Yazilim ve Bilgisayar Sistemleri | System & DevOps Intern

Aug 2024 - Oct 2024

- > Containerised development environments with Docker; configured Kubernetes deployment workflows for .NET and JavaScript projects.
- > Built automated CI/CD pipelines, reducing deployment and debugging cycle time by an estimated 60% on targeted services.
- > Implemented Linux server automation scripts and security hardening configurations.

CONSULTA Management and Technology Consulting | Technical Consultant (PLM Team) *Jun 2023 - Sep 2023*

> Delivered database configuration, RESTful API integration, and system integration support for Centric PLM across corporate clients — hands-on technology consulting delivery to enterprise stakeholders.

S E L E C T E D P R O J E C T S

Financial AI & RAG Systems

Financial RAG LLM System - Document Intelligence Pipeline - *Python, LangChain, LlamaIndex, ChromaDB, pgvector, Embeddings, FastAPI*

- > End-to-end RAG pipeline for large-scale financial document processing: PDF ingestion, text chunking, embedding-based semantic indexing (ChromaDB + pgvector), and dynamic querying via LangChain and LlamaIndex orchestration.
- > Architecture supports multi-document retrieval with context window management — enabling accurate Q&A and decision support over dense financial corpora: regulatory filings, compliance documents, financial reports.
- > Modular design with swappable embedding models and LLM backends (Gemini, Ollama, vLLM) — production-ready architecture applicable to regulatory reporting, risk analysis, and financial document intelligence workflows.

Document AI Pipeline - OCR, RAG & Multi-Agent Analysis - *OCR, LangChain, Gemini, ChromaDB, pgvector, MCP, Multi-Agent, OnnxRuntime, Streamlit*

- > Multi-layered AI pipeline covering two modes: (1) conversational PDF analysis via LangChain (pypdf > text-split > ChromaDB embed > Gemini generate), OnnxRuntime embeddings, Streamlit interface; (2) multi-document OCR ingestion > pgvector semantic indexing > MCP-based multi-agent orchestration — reducing document retrieval from hours to under a minute.
- > Multi-agent coordination handles routing, retrieval, and response synthesis — applicable to regulatory reporting, compliance document review, and financial audit workflows.

Kentas AI Asistani - Enterprise Local LLM Deployment - *vLLM, WSL2, GPU Inference, Fine-Tuning Pipeline, FastAPI*

- > Institution-wide AI assistant deployed on local infrastructure: vLLM inference server (WSL2 + GPU), custom system instructions optimised for domain-specific queries, API security layer, stress-tested for production load.
- > Autonomous fine-tuning data pipeline captures and labels production query-response pairs via better-sqlite3 — structured retraining dataset without manual intervention, designed for continuous model improvement.

TalepAI - Multi-LLM Municipal AI Platform - *Gemini, vLLM, Ollama, Next.js, SQLite Cache, Rate Limiting, Playwright*

- > Production AI platform with runtime-switchable LLM backends (Gemini/OpenAI-compatible/Ollama/vLLM) classifying requests into structured JSON (category, urgency, confidence score, recommended action).
- > SQLite result caching, 13s rate-limit management, 60s auto-backoff on 429 responses, batch processing at 50 requests/cycle with retry. Full Playwright e2e + Jest unit test suite.

ML & Data Engineering

ParselAI - Hybrid Rule-Based + ML Analysis Engine - *FastAPI, XGBoost, LightGBM, CatBoost, Optuna, PostGIS, ARQ, Redis*

- > Hybrid rule-based + ML ensemble: structured compliance rule engine + XGBoost/LightGBM/CatBoost ensemble with Optuna hyperparameter optimisation. Async FastAPI + ARQ Redis queue for batch ML inference at scale; WeasyPrint PDF report generation.
- > PropTech platform automating Istanbul parcel buildability analysis: TAKS/KAKS/setback rules compliance engine + ML ensemble. Architecture pattern applicable to credit scoring, risk assessment, and regulatory compliance scoring.

AI-Assisted MOSFET Amplifier Design - *CNN, Python, PyTorch, LTspice, Automated Data Generation*

- > CNN model trained on 98,000 automatically generated LTspice simulation data points, predicting optimal analog circuit parameters — end-to-end ML pipeline benchmarked against expert manual output.

AI Strategic Board Game - *Reinforcement Learning, Q-Learning, React*

- > AI agent making complex strategic decisions using Q-Learning reinforcement learning; integrated with a React web interface.

GIS / Urban Tech

GezelimS / Gezelim App - Istanbul Discovery Platform - *Published, Firebase, React Three Fiber, MapLibre, proj4, Playwright, AI Data Pipeline, Google POI*

- > Live Istanbul exploration platform: Firebase 12 (Firestore realtime + Auth), React Three Fiber + MapLibre GL hybrid, proj4 coordinate projection (EPSG transforms), TanStack Virtual windowed rendering for large datasets.
- > AI data scrubbing pipeline: Playwright-based API interception pulling and validating Google POI data, AI-assisted quality control and deduplication at scale, dynamic routing and gamification mechanics on the validated dataset.

ParselInfo360 - GPU-Accelerated Parcel Platform - *Deck.gl, luma.gl, WebGL2, Leaflet, Express, Team Project*

- > Neighbourhood parcel platform with Deck.gl 9 + luma.gl WebGL2 GPU-accelerated 3D geospatial rendering, hybrid Leaflet/Deck.gl map layers, Express + multer report upload API, Radix UI design system, admin marketplace panel.

EUPro - EU Grant Calls Automated Scraper - *Playwright, GitHub Actions, Google Geocoding, Node.js, CI/CD*

> Automated data pipeline for EU funding calls: Playwright scraping, Google Geocoding API enrichment, Turkish translation pipeline, GitHub Actions for scheduled auto-update with zero manual intervention.

Full-Stack & Web Applications

3D / WebGL / XR

Digital Showroom XR Ecosystem - *Unreal Engine C++, Three.js, React Three Fiber, WebXR, Pannellum, Next.js, Zustand, GSAP, iOS/Android*

- > Real-Time Interactive Simulation (Unreal Engine + C++): first-person & VR-capable high-performance simulation with dynamic material configuration, real-time design switching, and custom movement mechanics — deployable on PC and VR headset.
- > Web-Based Spatial CMS & 3D Scene Editor (Three.js + Next.js + Zustand): Matterport-style dual-layer system — editor for 3D scene creation, hotspot management, lighting and post-processing (AO/bloom/vignette/colorGrading); full undo/redo history, GSAP animations, JSZip scene export/import.
- > Cross-Platform WebXR & AR Pipeline: browser-based panoramic tours (Pannellum, SVG radar minimap with yaw tracking, ambient + per-scene audio, Vercel Blob storage) and WebAR for iOS/Android with gyroscope permission handling, inside-out panorama technique, and lerpFactor rotation smoothing.
- > Custom Gaze-Based Panotour Engine (Vanilla JS + Three.js): original panoramic tour engine with VR headset gaze-interaction built entirely from scratch — no external tour libraries.

Erim'in Odası - 3D WebGL Portfolio + AI Chatbot - *Published, React Three Fiber, Groq LLaMA 3.3-70b, WebGL Fluid, BVH*

- > Live interactive 3D portfolio: WebGL fluid simulation background, three-mesh-bvh accelerated ray casting, Groq LLaMA 3.3-70b AI chatbot (13 projects defined, fuzzy matching, MAX_HISTORY=12), Playwright A/B performance testing. Live at erimkun.github.io/erimodasi.

DevTools / Open Source

Claude Copilot Bridge - *Published, VS Code Extension, Multi-Agent Orchestration, TypeScript*

- > Published VS Code extension connecting Claude Code to GitHub Copilot Chat as an autonomous subagent over a local HTTP bridge (127.0.0.1:54321) — 100% local, no central server. Available on VS Code Marketplace.

Hibeci Agent - EU Grant Writing AI Agent - *GitHub Copilot Agents, Logical Framework, Horizon Europe, IPA*

- > GitHub Copilot agent automating EU grant proposal writing (Horizon Europe, IPA) using Logical Framework methodology, 5N3K activity structure, anti-hallucination strict mode, pdf-extraction and memory-merger skills.

E D U C A T I O N

Yeditepe University - B.Sc. Electrical & Electronics Engineering (English, Scholarship)	2021 - 2025
Istanbul University - Associate Degree, Web Design & Coding	2023 - 2026
Ataturk University - Bachelor's, Business Administration (Open Learning)	2025 - Present
ISTEK Semiha Sakir Anatolian High School - Graduated 2nd in school	2016 - 2021

C E R T I F I C A T I O N S

Unreal Engine XR Development (Epic Games) | AR/VR/MR/XR with WebXR (University of Michigan) | C++ for Unreal (University of Colorado)

Deep Learning with PyTorch | Neural Networks and Deep Learning | Computer Vision with YOLOv11 | Project Management Foundations

L A N G U A G E S

English C1 Advanced (academic & technical proficiency) **German** A2 **Spanish** A1